

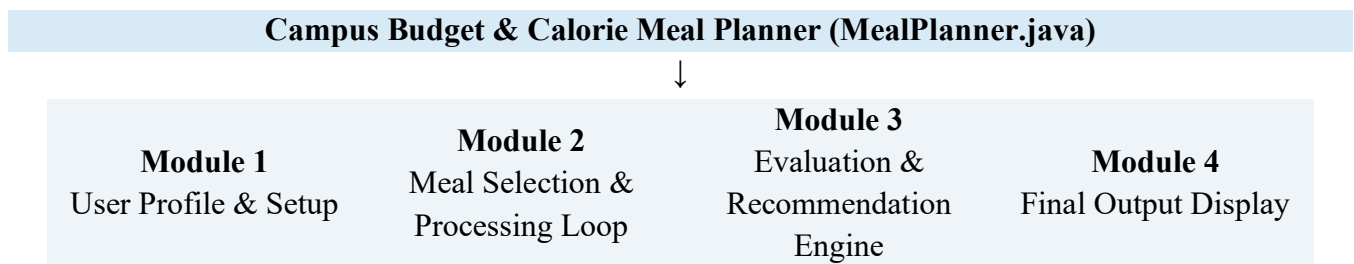
Modular Program Hierarchy

Campus Budget & Calorie Meal Planner

Purpose of the Program Hierarchy

The Campus Budget & Calorie Meal Planner can be structured into four modules. In the current version, most of these functions are implemented within the main () method, but they represent separate responsibilities that could later be refactored into individual methods.

Overall Program Hierarchy



Module 1: User Profile & Setup

Role: Collects the information needed before meal selection begins and establishes the user's daily calorie goal.

Main responsibilities:

- Displays the welcome message with JOptionPane.
- Prompts the user to enter a daily meal budget.
- Converts the budget from String input to a double value.
- Prompts the user to select one of the provided gender options to determine the calorie target.
- Sets the calorie target to 2,000 or 2,500 calories and initializes total calories at zero.

Related Java code:

```
JOptionPane.showMessageDialog(null, "Welcome to the Campus Budget & Calorie Meal Planner!");  
String budgetInput = JOptionPane.showInputDialog("Enter your daily meal budget ($):");  
double remainingBudget = Double.parseDouble(budgetInput);  
int calorieTarget = (genderChoice == 0) ? 2000 : 2500;  
int totalCalories = 0;
```

Input

User-entered daily budget and gender selection.

Processing

Converts the budget to a numeric value and assigns the correct calorie target.

Output

Starting budget, daily calorie target, and an initial calorie total of 0.

Module 2: Meal Selection & Processing Loop

Role: Allows the user to choose meals repeatedly while the program updates both the remaining budget and total calories.

Main responsibilities:

- Displays Breakfast, Lunch, Dinner, and Finish & View Summary options.
- Uses a while loop so the user can select more than one meal.
- Subtracts the selected meal price from remainingBudget.
- Adds the selected meal calories to totalCalories.
- Ends the loop when the user chooses the finish option.

Related Java code:

```
while (ordering) {  
    int selection = JOptionPane.showOptionDialog(...);  
    if (selection == 0) {  
        remainingBudget -= BREAKFAST_PRICE;  
        totalCalories += BREAKFAST_CALORIES;  
    } else if (selection == 1) { ... }  
}
```

```

        else if (selection == 2) { ... }
        else { ordering = false; }
    }

```

Input

Breakfast, lunch, dinner, or finish selection.

Processing

Uses fixed meal constants to subtract cost and add calories after every selection.

Output

Updated remaining budget and updated total calorie count.

Module 3: Evaluation & Recommendation Engine

Role: Compares the user's total calories with the calorie target and creates a simple recommendation based on the result.

Main responsibilities:

- Checks whether totalCalories is less than or equal to calorieTarget.
- Provides a positive message when the user stays within the target.
- Calculates excess calories when the user goes over the target.
- Estimates jogging distance using approximately 100 calories per mile.
- Uses Math.ceil() so a partial mile is rounded up to the next whole mile.

Related Java code:

```

if (totalCalories <= calorieTarget) {
    summary += "Great job! You stayed within your healthy recommended calorie balance.";
} else {
    int excessCalories = totalCalories - calorieTarget;
    int milesToJog = (int) Math.ceil((double) excessCalories / 100);
    summary += "Suggested Workout: Consider jogging " + milesToJog + " mile(s)";
}

```

Input

Total calories consumed and the assigned calorie target.

Processing

Compares totals, calculates excess calories if necessary, and converts excess calories to suggested jogging miles.

Output

Calorie-status message and, when needed, a suggested jogging distance.

Module 4: Final Output Display

Role: Formats the final financial and nutritional information and presents it to the user through a JOptionPane dialog box.

Main responsibilities:

- Builds one summary String containing the final results.
- Formats the remaining budget to two decimal places.
- Displays total calories consumed and the target goal.
- Appends either the success message or workout recommendation.
- Shows the completed summary in a final information dialog.

Related Java code:

```
String summary = "=== Daily Summary ===\n" +  
    "Remaining Budget: $" + String.format("%.2f", remainingBudget) + "\n" +  
    "Total Calories Consumed: " + totalCalories + " cal\n" +  
    "Target Goal: " + calorieTarget + " cal\n\n";  
JOptionPane.showMessageDialog(null, summary, "Final Results",  
    JOptionPane.INFORMATION_MESSAGE);
```

Input

Remaining budget, total calories, calorie target, and recommendation text.

Processing

Formats all information into a readable final summary.

Output

A final JOptionPane window containing the user's financial and nutritional results.

How the Modules Work Together

Module 1 gathers the starting information. Module 2 handles meal choices and updates the running totals. Module 3 evaluates the final calorie amount and creates a recommendation. Module 4 formats and displays all results. This structure supports the project purpose because the program tracks budget and calories at the same time while keeping the user interaction quick and simple.

Modular Design Summary

Module	Primary Responsibility	Key Variables / Features
1. User Profile & Setup	Collects budget and sets calorie target.	budgetInput, remainingBudget, genderChoice, calorieTarget
2. Meal Selection & Processing	Processes meal choices and updates totals.	ordering, selection, meal price/calorie constants
3. Evaluation & Recommendation	Checks calorie goal and calculates workout suggestion.	totalCalories, excessCalories, milesToJog, Math.ceil()
4. UI Output Display	Formats and displays final results.	summary, String.format(), JOptionPane